

# JAVIER REBULL, M.Sc.

Senior Full Stack Engineer | AI & ML Specialist | Software Architect

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## Summary

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Senior Full Stack Engineer and AI/ML Specialist with 20 years of experience in banking technology and enterprise software engineering, delivering mission-critical solutions across financial services, payments, credit cards, and large-scale platforms, from discovery and technical scoping through system architecture, build, and production rollout. Hands-on in deploying machine learning models to production, building MLOps pipelines, and leading large-scale platform migrations without critical incidents, aligning executives, engineering teams, risk partners, and external banks around measurable outcomes.

## Technical Skills

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**Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, COBOL, SAS

**AI / Machine Learning:** Deep Learning, NLP, Time Series Forecasting, Computer Vision, Generative AI, LLMs, MLOps

**Frameworks & Libraries:** PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Prophet, DeepAR, React, Angular

**Cloud & DevOps:** AWS (SageMaker, Lambda, S3, EC2), Cloud Architecture, Mainframe-to-Cloud Migration, CI/CD, Docker, Linux, Git/GitHub

**Software Engineering:** Full Stack, System Architecture, Distributed Systems, Microservices, REST/SOAP, Agile/Scrum

**Databases:** Oracle, DB2, MySQL, MongoDB

**Financial Technology:** Core Banking, Payments, Credit Cards, Loans, Regulatory (FDIC 370), High-Availability Systems

## Work Experience

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**Senior Associate Application Development** (since Mar 2024; Associate Application Development, Sep 2019 – Feb 2024)

Santander Bank US — Quincy, MA

Sep 2019 – Present

- Led the Gravity Middleware Project end to end, migrating IBM Mainframe/DB2 to private cloud (Microfocus/Oracle) with zero critical incidents, and moved 500+ WAS/WebSphere servers to modern cloud infrastructure.
- Built end-to-end MLOps pipelines for automated model deployment and monitoring in production banking environments.
- Directed full stack development of Certificates of Deposit for OpenBank, from front-end interfaces to back-end services.
- Delivered deadline-driven regulatory initiatives: SBNA Loans COVID-19 Relief Program and Regulatory FDIC 370.
- Designed UI/UX for internal banking applications, raising user satisfaction by 35%.
- Kept distributed banking systems incident-free through root-cause analysis across hardware, network, and service layers, and led design and code reviews on testability and code quality.

**Computer Systems Analyst**

Vass IT US — Cambridge, MA

Apr 2013 – Sep 2019

- Built Computer Vision solutions for document processing and fraud detection, integrating ML models into client systems.
- Led full stack development of loan-management applications and managed IT project portfolios and PMO activities.
- Owned software products from requirements through deployment and maintenance for client business goals.

**Spec Analytics Analyst 2**

Citibank Banamex — Mexico City, Mexico

Jun 2010 – Apr 2013

- Designed and optimized secure distributed systems for global financial services, processing millions of daily transactions with high availability.
- Built data-driven algorithms (SAS/Python) that raised marketing response rates 20% and cut delinquency 10%.
- Built interactive dashboards and front-end interfaces for executive reporting and real-time data visualization.

**Senior Programmer**

Accenture — Mexico City, Mexico

Aug 2007 – May 2010

- Developed and maintained large-scale core financial systems (COBOL, Java) for client banks BBVA Bancomer and Santander México, with zero critical incidents.
- Designed front-end user interfaces for mainframe applications, improving operational efficiency and developer experience.

## Systems Analyst & IT Help Desk Analyst

Microsistemas / Ford Credit Company — Mexico City, Mexico

Aug 2006 – Aug 2007

- Help desk consultant for Ford's department: system stability, security, and software and hardware support.

## Education

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### Universidad Autónoma de Ciudad Juárez (UACJ)

Ciudad Juárez, Mexico

M.S. in Artificial Intelligence and Data Analytics | Expected 2027

Focus: Smart Optimization, Linear Optimization, Non-Linear Optimization, Data Engineering

### Tecnológico de Monterrey (ITESM)

Mexico City, Mexico

M.Sc. in Applied Artificial Intelligence | Conferred 2026

GPA: 4.0/4.0 • Coursework: Data Science, Machine Learning, AI, Data Visualization, Fintech & IoT

### The National Polytechnic Institute of Mexico (ESCOM IPN)

Mexico City, Mexico

B.S. in Computer Science & Electronics Engineering | 2006

GPA: 3.28/4.0 • Coursework: Programming, Software Engineering, AI, Networks, Data Structures, Algorithms & OS

## AI / ML Projects

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- **EpiForecast-MX — Epidemiological Forecasting Platform** (2025–2026): M.Sc. capstone at Tecnológico de Monterrey, developed in collaboration with the Instituto Mexicano del Seguro Social (IMSS). Forecasts weekly notified cases across Mexico's 32 federal entities on a 52-week horizon, selecting per series among Prophet, DeepAR, Ensemble and Stacking models through an auditable rule. Deployed in production at [epiforecast.mx](https://epiforecast.mx).
- **Text Classification with RNNs — AG NEWS Dataset** (Mar 2025): Built an RNN in PyTorch to classify AG NEWS articles, covering data preprocessing, model architecture design, training pipelines, and evaluation.
- **IoT for Sustainable Urban Development** (Feb 2025): Investigated IoT solutions for traffic congestion in Boston and public lighting efficiency in Mexico City using distributed sensor networks.
- **Multilayer Fully Connected Network in NumPy** (Jan 2025): Implemented a neural network from scratch for ASL letter recognition, including forward/backward propagation and performance benchmarking.
- **Stock Price Prediction for Apple** (Nov 2024): Developed and compared LSTM, ARIMA, Prophet, and Naïve predictive models on historical market data using advanced statistical methods.

## Conferences & Publications

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- **Oral presentation — CALASS 2026, Congrès de l'ALASS**, Université de Montréal, Canada (Aug 27–29, 2026): *“De la surveillance épidémiologique à l'intelligence prédictive : le rôle de l'intelligence artificielle dans l'organisation prospective des systèmes publics de santé.”* Presented EpiForecast-MX (Tecnológico de Monterrey & IMSS).
- **ePoster — 26th WPA World Congress of Psychiatry**, Stockholm, Sweden (Sep 23–26, 2026), Digital Psychiatry track: *“Population-level prediction of depression, Alzheimer's disease, and Parkinson's disease incidence using artificial intelligence: implications for health service planning.”*
- **Poster — 2nd Public Health Conference & 2nd HSG Europe Preconference**, Skopje, North Macedonia (May 21–22, 2026): *“AI-Based Forecasting to Shift Epidemiological Surveillance toward Predictive Approaches: Depression, Alzheimer's, and Parkinson's as Use Cases.”*
- **Article — Medium: Forecasting Dengue in Mexico: An End-to-End Data Science Story**. End-to-end data science article covering the full workflow for forecasting dengue cases in Mexico.

Certificates: [rebull.org/constancias](https://rebull.org/constancias)

## Training & Certifications

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Python Essentials 1 & 2 — Cisco (Jul 2026) • Lifelong Learning — Certiprof (Jan 2026) • AWS Certified AI Practitioner (Dec 2025) • Scrum Master Professional — SMPC® (Jun 2025) • Generative AI in Action — IBM SkillsBuild (Jun 2025) • AWS Certified Cloud Practitioner (Jan 2025)